Assignment-1

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Aim:
Implement basic string operations such as length calculation, copy, reverse, and concatenation using character single dimensional arrays without using built-in string library functions.

Objective:
- To implement basic string operations using one-dimensional character arrays.
- To calculate the length of a string without using built-in string functions.
- To copy, reverse, and concatenate strings manually using loops.
- To understand how character arrays store and manipulate strings in C++.

What I Learned.

Theory:
A string is a collection of characters stored in a one-dimensional character array and terminated by a special null character ('\0'). In C++, string operations can be performed either using built-in string library functions or manually using character arrays. In this program, all string operations are implemented without using any built-in string functions.

The program accepts two strings from the user and performs the following operations:

1. Length Calculation:

The length of each string is calculated by traversing the character array until the null character ('\0') is found. A counter is used to count the number of characters.

2. String Copy:

The first string is copied into another character array by copying each character one by one until the null character is reached.

3. String Reverse:

The first string is reversed by reading its characters from the last index to the first index and storing them in another array.

4. String Concatenation:

The two input strings are joined together by first copying the first string into a new array and then appending the second string at the end. Finally, the null character is added to complete the concatenated string.

Algorithm:

Algorithm 1: Length Calculation

1. Start.
2. Declare a character array to store the string.
3. Initialize a variable length = 0.

4. Read the string from the user.
5. Repeat until the null character ('\0') is found: -> Increment length by 1.
6. Display the length of the string.
7. Stop.

Algorithm 2: String Copy

1. Start.
2. Declare two character arrays: source string and destination string.
3. Read the source string.
4. Initialize index i = 0.
5. Repeat until the null character ('\0') is found:

->Copy source[i] to destination[i].

->Increment i.

6. Store the null character ('\0') at the end of the destination string.
7. Display the copied string.
8. Stop.

Algorithm 3: String Reverse

1. Start.
2. Read the input string.
3. Find the length of the string.
4. Initialize j = 0.
5. Traverse the string from the last character to the first character.
6. Store each character into the reverse array.
7. Append the null character ('\0') at the end of the reverse string.
8. Display the reversed string.

9. Stop.

Algorithm 4: String Concatenation

1. Start.
2. Read the first and second strings.
3. Copy all characters of the first string into the concatenated array.
4. Copy all characters of the second string after the first string.
5. Append the null character ('\0') at the end.
6. Display the concatenated string.
7. Stop.

Code:

```
#include <iostream>
using namespace std;

int findLength(char str[])
{
    int len = 0;
    while (str[len] != '\0')
    {
        len++;
    }
    return len;
}

void copyString(char str[], char copy[])
{
    int i = 0;
    while (str[i] != '\0')
    {
```

```
        copy[i] = str[i];
        i++;
    }
    copy[i] = '\0';
}

void reverseString(char str[], char reverse[])
{
    int len = findLength(str);

    for (int i = 0; i < len; i++)
    {
        reverse[i] = str[len - 1 - i];
    }

    reverse[len] = '\0';
}

void concatenateString(char str1[], char str2[], char concat[])
{
    int i = 0, j = 0;

    while (str1[i] != '\0')
    {
        concat[i] = str1[i];
        i++;
    }

    while (str2[j] != '\0')
    {
        concat[i] = str2[j];
        i++;
        j++;
    }

    concat[i] = '\0';
}

int main()
{
```

```
    char str1[20], str2[20];
    char copy[20], reverse[20], concat[20];
    int choice;

    cout << "Enter First String: ";

    cin>>str1;

    cout << "Enter Second String: ";

    cin>>str2;
    do
    {
        cout << "\n===== STRING OPERATIONS MENU =====";
        cout << "\n1. Find Length";
        cout << "\n2. Copy String";
        cout << "\n3. Reverse String";
        cout << "\n4. Concatenate Strings";
        cout << "\n5. Exit";
        cout << "\nEnter your choice: ";
        cin >> choice;

        switch (choice)
        {
        case 1:
            cout << "\nLength of First String = " << findLength(str1);
            cout << "\nLength of Second String = " << findLength(str2) <<
endl;
            break;

        case 2:
            copyString(str1, copy);
            cout << "\nCopied String = " << copy << endl;
            break;

        case 3:
            reverseString(str1, reverse);
            cout << "\nReversed String = " << reverse << endl;
            break;
```

```
        case 4:
            concatenateString(str1, str2, concat);
            cout << "\nConcatenated String = " << concat << endl;
            break;

        case 5:
            cout << "\nExiting Program...";
            break;

        default:
            cout << "\nInvalid Choice! Please try again.";
        }

    } while (choice != 5);

    return 0;
}
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\LOQ\OneDrive\Desktop\SY Preparation\college notes\DS\Assignmentsss>  & 'c:\Users\LOQ\.vsc
ode\extensions\ms-vscode.cpptools-1.32.2-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--std
in=Microsoft-MIEngine-In-jvmgtpgk.adc' '--stdout=Microsoft-MIEngine-Out-pqn51yuq.ggh' '--stderr=Micro
soft-MIEngine-Error-tysmawtx.ntc' '--pid=Microsoft-MIEngine-Pid-4hjkki3f.ca2' '--dbgExe=C:\mingw64\bi
n\gdb.exe' '--interpreter=mi'
Enter First String: samiksha
Enter Second String: adake

===== STRING OPERATIONS MENU =====
1. Find Length
2. Copy String
3. Reverse String
4. Concatenate Strings
5. Exit
Enter your choice: 1

Length of First String = 8
Length of Second String = 5
```

```
===== STRING OPERATIONS MENU =====
1. Find Length
2. Copy String
3. Reverse String
4. Concatenate Strings
5. Exit
Enter your choice: 2

Copied String = samiksha

===== STRING OPERATIONS MENU =====
1. Find Length
2. Copy String
3. Reverse String
4. Concatenate Strings
5. Exit
Enter your choice: 3

Reversed String = ahskimas
```

```
===== STRING OPERATIONS MENU =====
1. Find Length
2. Copy String
3. Reverse String
4. Concatenate Strings
5. Exit
Enter your choice: 4

Concatenated String = samikshaadake

===== STRING OPERATIONS MENU =====
1. Find Length
2. Copy String
3. Reverse String
4. Concatenate Strings
5. Exit
Enter your choice: 5

Exiting Program...
PS C:\Users\LOQ\OneDrive\Desktop\SY Preparation\college notes\DS\Assignmentsss>
```

Conclusion:

The program successfully implements the basic string operations such as length calculation, copy, reverse, and concatenation using one-dimensional character arrays without using built-in string library functions. It helps in understanding the internal representation of strings, character manipulation, array traversal, and manual implementation of string operations in C++.